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ICE 4.1



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(Each question is marked 2)1. Consider the problem of sharing secret keys that student A uses to exchange messages with his classmates, B, C, ..., Z. Of course, the key used by x to send messages to A is the same A uses to send messages to x. Note, messages shared between X and Y, for instance, are not shared with others. How many keys need to be established if everyone wishes to exchange messages with every one else?

- 1
- 26
- 26^2
- **325 (correct)**
- None of the above

2. There are two possibilities for using electronic means to establish a session key between, for instance, X and Y:

(A) X delivers the key to Y after suitably encrypting it and sending it over a pre-established secure channel between X and Y; or

(B) using a trusted third party, call it ABC, to generate and deliver the session key after suitably encrypting it and sending the same to both X and Y using existing secure channels to each X and Y. Which one would you prefer? Method A or B, or ...?

- method A, since a third party is not required
- **method B, since there is no need for pre-established secure channel between every pair of entities, X and Y (correct)**
- method B, but only when there exists a secure channel between X and Y for all X, Y
- None of the above

3. What is the purpose of including a $f(N1) = N1+1$ in a response to a message that contained a nonce $N1$?

- It is simply a nonce, nothing more
- **It helps relate the response to the message that contained $N1$ (correct)**
- It permits one to compute $N1$, given $N1$
- None of the above

4. In what way does it help to establish a (new) session key, K_s between entities X and Y, when there already exists a master key, K_m between X and Y.

- The master key may be used for long period of time or for short period of time



- **Helps establish a session that is expected to last a much shorter duration (correct)**
- It does not seem necessary at all
- None of the above

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